Differential object marking in the L2 Spanish of L1 German speakers: A corpus-based analysis

Philippa Adolf
(Katholische Universität Eichstätt-Ingolstadt)

El presente estudio contribuye a la investigación de la adquisición del marcado diferencial de objeto (ingl. *differential object marking*, DOM) del español como segunda lengua (L2), evaluando cuantitativamente el uso escrito del DOM por hablantes nativos de español europeo y por parte de hablantes de alemán como lengua primera (L1). Mediante un enfoque basado en corpus, se analizan más de 100 textos escritos en términos de la alternancia *a*/∅ con objetos directos nominales, evaluando un conjunto de predictores semánticos y de estructura de la información utilizando un análisis multifactorial basado en clasificaciones de bosques aleatorios. Con este enfoque se investiga hasta qué punto y en función de qué factores el grupo de hablantes L2 difiere del grupo nativo en su uso del DOM. Los resultados indican que los aprendices son capaces de utilizar el DOM de manera similar al hablante nativo, pero omiten una cantidad considerable de marcadores en casos de objetos directos animados en los que los hablantes nativos usarían el DOM.

Differential object marking, L1 German speakers, written texts, target-likeness, random forest analysis

1. Introduction

The use of the marker *a* with direct objects in Spanish is a well-known instance of the general phenomenon of Differential Object Marking (DOM) (Parodi/Avram 2018: 63). Originally introduced by Bossong (1991), DOM refers to a phenomenon whereby a subset of direct objects (DO) are differentially marked.

(1) a. Encontré un problema.
found a problem$_{DO}$
'I found a problem.'
b. Encontré a un superviviente.
found DOM a survivor$_{DO}$
'I found a survivor.' (Fábregas 2013: 2)

The absence of the marker *a* in example (1a) compared to (1b) points to an underlying, not exclusively syntactic difference between these two structures. DOM has been shown to be dependent upon additional factors, including the semantic characteristics and the information structure of the noun phrase, and the influence of the verb appearing in the DOM construction, which renders DOM an overall complex phenomenon (cf. Kabatek et al. 2021). This complexity has been shown to negatively impact the acquisition process for L2 speakers of Spanish, where DOM is considered a particularly vulnerable domain (Ciovarnache/Avram 2013; Ticio

2015). The investigation of DOM has brought forth both experimental studies (Bowles/Montrul 2009; Montrul/Sánchez-Walker 2013; Nediger et al. 2016; Pomino et al. 2018; Montrul/Mardale 2020) and a small amount of corpus studies (von Heusinger/Kaiser 2011; Heredero 2021; Simon 2022). To my knowledge, no corpus study has yet been conducted on the use of DOM in the L2 Spanish spoken by L1 German speakers. To answer the questions of whether learners deviate from natives in their use of DOM, and if so, in which ways, parts of the L1 and L2 subcorpora of the CEDEL2 (*Corpus escrito del español como L2*, Lozano 2022) corpus will be analyzed using a novel statistical approach called MuPDARF (Multifactorial Prediction and Deviation Analysis using Random Forests; see 4.2). The purpose of this study is therefore to provide usage-based evidence for a better insight into how L1 German learners of Spanish use DOM. The results may further inform foreign-language didactics with respect to instructional methods of Spanish DOM. The structure of this paper is as follows: Section 2 presents previous research on the intricacies of Spanish DOM and the challenges in the L2 acquisition of DOM. Based on previous research into the L2 acquisition of DOM, a set of hypotheses is introduced in Section 3, followed by details on the used corpora, the composition of the data, and the applied statistical methods in Section 4. Section 5 presents the results of the analysis, which will be discussed in Section 6. Section 7, finally, offers a conclusion.[1]

2. Background

Providing an overview of the extensive research conducted on Spanish DOM within the relatively restricted boundaries of a single paper is not an easy task (cf. Kabatek et al. 2021 and von Heusinger 2008 for an overview). The descriptions of the following section will therefore not aim to be exhaustive but rather attend to those parameters that are relevant to this study.

2.1 Identified factors and intricacies of Spanish DOM

In the past, DOM has often been considered a syntactic phenomenon (Fábregas 2013: 30). The motivation for this can be derived from the observation that DOM seems to be obligatory in certain structures, such as in combination with personal

[1] This study's data and background section is based on my bachelor's thesis *A corpus-study on the use of DOM in the L2-Spanish of L1-German speakers* (2023). The analysis is however a new one, using a different statistical method.

pronouns (2), and blocked in others, for instance in ditransitives due to the identical morphological shape of indirect objects (IO) (3).

(2) (Te) vi *(a) ti.
(you$_{DO}$) saw *(DOM) you$_{IO}$
'I saw you.' (Fábregas 2016: 30)

(3) Ana present *a María a Juan.
Ana presented *DOM María$_{DO}$ Juan$_{IO}$
'Ana introduced María to Juan.' (Ticio 2015: 66)

The occasional *a*-marking of direct objects was originally assumed to be primarily used on those objects which could easily be confused for subjects (SBJ) if not overtly marked (Jegerski 2018: 587), thus trying to further explain Spanish DOM as a phenomenon which holds a discriminatory function in a language with relatively flexible word order. Example (4a) shows an unmarked order of constituents in a sentence containing a direct object. Contrastingly, the roles are reversed in (4b) where Ana can now be identified as the direct object and Pedro as the subject of the sentence while maintaining the original order of constituents. The difference appears to be dependent upon the use of *a*. Note that the second example represents a marked word order in which the fronted direct object is focalized.

(4) a. Ana sigue a Pedro.
Ana$_{SBJ}$ follows DOM Pedro$_{DO}$
'Ana follows Pedro.'

b. [$_F$ A Ana] sigue Pedro.
DOM Ana$_{DO}$ follows Pedro$_{SBJ}$
'Pedro follows Ana.' (Jegerski 2018: 587)

However, DOM cannot only be related to case marking since one of the necessary characteristics of syntactically motivated case marking is that it should be compulsory in all respective contexts. As can be inferred from the two grammatical utterances of examples (1a) and (1b), this is not how Spanish DOM works, which means that DOM is not only regulated by syntactic factors but also by additional parameters. For a better insight, these parameters will be presented according to their respective structural levels in the following subsections.

2.1.1 Semantic factors

One important characteristic of Spanish DOM is that there seem to be two interacting semantic scales which rank the probability of whether *a*-marking should be applied or not: one for animacy, and one for specificity (Fábregas 2013: 10). As for animacy, with the data of the upcoming analysis in mind, the commonly made distinction between [±animate] and [±human] will not be relevant for the outcome of the study and will therefore be disregarded. Specificity shares similarities with the concept of *definiteness* (e.g., both expressed in the form of determiners, pronouns, or the lack thereof) but differs slightly in its semantic interpretation in that it focusses on the uniqueness of an entity in a given context (Lambrecht 1994: 79). For this study, definiteness and specificity can be interpreted as largely congruent concepts, a view which has been adopted from Ticio/Avram (2015).

Much work has already focused on the relationship between these semantic features and DOM and the way they interact with each other (García García 2005; von Heusinger 2008; Pomino 2012). Out of the two, animacy has generally been considered the more dominant one (cf. example 1), to the point where *a*-marking has been described to be only acceptable with animate noun phrases and ungrammatical with inanimate ones, regardless of other semantic conditions (García García 2005: 23). Still, *a*-marking is found with inanimate direct objects as well. In (5), the marker *a* occurs with the inanimate direct object *sustantivo*.

(5) El adjetivo modifica al sustantivo.
the adjective modifies DOM-the noun$_{\text{DO [–ANI, +SPEC]}}$
'The adjective modifies the noun.' (García García 2005: 64)

Moreover, animate direct objects can also be left unmarked in some contexts, yielding a non-specific reading (6a). While the sentence without the *a*-marked direct object suggests that José needs a doctor in general, the first version of the sentence in (6b) implies that José might already have a specific kind of doctor or a specific person in mind.

(6) a. José necesita un médico.
José needs a doctor$_{\text{DO [+ANI, –SPEC]}}$
'José needs a doctor.'
b. José necesita a un médico.
José needs DOM a doctor$_{\text{DO [+ANI, +SPEC]}}$
'José needs a doctor.' (Bowles/Montrul 2009: 200)

In an effort to integrate animacy and specificity into a system that explains the contexts in which DOM occurs, individual scales for animacy and specificity have been proposed by Ticio/Avram (2015), Pomino (2012), and von Heusinger (2008). A possible synthesis of the individual scales can be found in (7), where both factors are subsumed in a single linear structure called Referentiality Scale, with the expectation of DOM decreasing from left to right.

(7) Referentiality Scale:
PERSONAL PRONOUN > PROPER NOUN > DEFINITE NP > INDEFINITE NP > NON-SPECIFIC NP
(von Heusinger 2008: 5)

The prototypical cases of DOM seem to clearly align with this distribution. It therefore appears that direct objects that are higher on the animacy and specificity scale are more likely to be overtly differentially marked with *a*.

2.1.2 Information-structural factors

Some studies of DOM have drawn attention to factors beyond the direct object NP and analyzed the influence of the topicality of a direct object on the use of DOM (Leonetti 2004; García García 2014; Kabatek 2016; Kabatek et al. 2021), pointing out that topicality shows high correlations with the referential dimensions of animacy and specificity since prototypical topics are usually specific and animate. According to Fábregas (2016), there are three notions that have been highlighted in what is taken to be, loosely, a topic, with each of them being accredited different importance by different authors: *givenness*, the contextual information of a topic of discourse is already known to the listener (cf. Matić 2015: 96); *aboutness*, topic as an expression whose referent the sentence is about (Lambrecht 1994: 118; Melis 2021: 47); and initial positioning (Lambrecht 1994: 117). The connection to DOM can be made from the observation that, cross-linguistically, DOM has been widely attested to be a strategy to mark topical objects (Bossong 1991; Mayer/Sánchez 2021). Instances of *a*-marked direct objects which are individuated as discourse topics are cases of object fronting (8a), and clitic left dislocation (8b). The fact that the use of *a* is obligatory in these cases supports the idea of a possible correlation between DOM and topicality as pointed out by Leonetti (2004) and Pomino (2012).

(8)	a.	[$_{TOP}$ *(A) Juana]	vi.	
		*(DOM) Juana$_{DO}$	saw	
		'I saw Juana.'		
	b.	[$_{TOP}$ *(A) Juana]	la	vi.
		*(DOM) Juana$_{DO}$	her	saw
		'I saw Juana.'		

(cf. Fábregas 2013: 48)

Another category relating to the topicality of a direct objects is *object continuity*, which refers to the extent to which constituents maintain their referential identity in continuous clauses (Paltridge 2012: 75). As such, object (dis)continuity is defined by the way in which an object is introduced and continued throughout the development of the discourse (cf. examples 9a, discontinuous, and 9b, continuous). With many continuous objects having been observed to be *a*-marked (Fábregas 2013: 33), object (dis)continuity should be considered in information-structural analyses of DOM.

(9) a. y pone *a*l bebé en él. La mujer descubre de nuevo
and puts DOM-the baby in it the woman discovers again

*a*l bebé en el carro. (L1, T02, 19–20)
DOM-the baby in the pram

'and he puts the baby in it. The woman discovers the baby again in the pram.'

b. Ve *a* una señora con un bebé en un carrito
sees DOM a woman with a baby in a pram

y deja *a*l bebé en el carrito. (L2 Spanish, T66, 17–18)
and leaves DOM-the baby in the pram

'He sees a woman with a baby in a pram and leaves the baby in the pram.'

In other words, speakers may resort to mark direct objects to indicate a high degree of discourse prominence – expressed in the form of givenness and continuity – that would otherwise naturally tend to be less prominent (Leonetti 2004: 23).

2.1.3 Transitivity-related factors

Many authors have pointed out that, apart from the inherent nominal factors and its discourse status, the properties of the verb also constitute a crucial factor impacting *a*/Ø-alternation and should therefore be included in analyses of DOM (García García 2014, 2018; Kabatek et al. 2021; Heredero 2022). Regarding these verbal factors, different sets of parameters are found throughout the literature. To encompass the majority of these factors within one general notion, von Heusinger (2008) proposes that the basic conditioning factor of DOM is *transitivity*, which is conceived to be a gradual property rather than an absolute one. With regard to DOM, the idea is therefore that *a*/Ø-alternation is a way of marking high or low transitivity, meaning that the more properties of the predicate and the direct objects coincide with those that are subsumed under the notion of high transitivity, the more likely it is that DOM is applied (Melis 2021). Out of the various parameters which influence transitivity, telicity and affectedness will be discussed in more detail, the reason being that these factors are the ones which have received the most coverage in the literature as pointed out by von Heusinger/Kaiser (2011).

As an aspectual category, telicity is generally regarded as a property of the verb phrase that implies the existence or absence of an endpoint or culmination after which the event denoted by the predicate ends or is completed (Heredero 2021: 317, 2022: 94). In (10a), the omission of the marker *a* would be considered ungrammatical, whereas, in example (10b), *a*-marking seems to be optional. Following Torrego's (1999) view, this is due to the verb *saludar* being telic and the verb *observar* being atelic in nature.

(10) a. María saludó a/*∅ un compañero.
María greeted$_{[+\text{TEL}]}$ DOM/*∅ a colleague$_{\text{DO}}$
'María greeted a colleague.'
b. Observaron a/∅ un niño.
watched$_{[-\text{TEL}]}$ DOM/∅ a child
'They observed a child.' (Torrego 1999: 1789)

Krifka (1998) originally stresses that the aspectual properties of a predicate stem from two sources: the nature of the verbal nucleus and the nature of the argument and its additions. Consequently, not only the predicate but possible adjuncts will be considered for an accurate judgement.

Affectedness is another factor that has been mentioned as fundamental when it comes to the influence of the verb on the distribution of DOM. It can be understood as "the persistent change of an event participant" (Kabatek et al. 2021: 149). Tsunoda's and Hopper/Thompson's transitivity scales (11) propose a graded notion of affectedness, with transitive verbs being ranked according to the degree to which the participant is engaged or transformed (Tsunoda 1985: 388; von Heusinger/Kaiser 2011: 593). The empirical studies on affectedness and DOM conducted by the above-mentioned authors have found that the distribution of DOM in Spanish largely aligns with these scales in numerous cases. When applied to DOM, a decrease of *a*-marking is assumed to be observed from the first to the last group.

(11) a. Tsunoda (1985: 388)
ACTION > PERCEPTION > PURSUIT > KNOWLEDGE > FEELING
b. Hopper and Thompson (1980: 252)
DIRECT EFFECT ON PATIENT > PERCEPTION > PURSUIT > KNOWLEDGE > FEELING

direct effect on patient	**perception**	**pursuit**	**knowledge**	**feeling**
matar 'kill'	*mirar* 'watch'	*buscar* 'search'	*conocer* 'know'	*querer* 'want'
herir 'hurt'	*oír* 'hear'	*seguir* 'follow'	*entender* 'understand'	*temer* 'fear'
cambiar 'change'	*escuchar* 'listen'	*esperar* 'await'	*aprender* 'learn'	*amar* 'love'

Table 1. Examples of verbs along the transitivity scale, based on Wall/Obrist (2021: 142).

Due to the numerous factors the notion of transitivity entails, telicity and affectedness can be regarded as useful in describing the potential influence of the semantic properties of the verb on DOM. It appears as though the marker *a* aligns with many of the characteristics which are assigned to a high degree of transitivity, yielding a varied perspective on DOM as a means of marking transitivity.

2.2 Spanish DOM as a vulnerable domain in the L2 acquisition of L1 German speakers

With the numerous conditioning factors of Section 2 in mind, suffice it to say that Spanish DOM is a syntactically, semantically, and pragmatically complex phenomenon. While this complexity does not seem to negatively impact the acquisition process of DOM for L1 speakers (Rodríguez Mondoñedo 2008), this is generally not the case for L2 acquisition of Spanish, where DOM is considered particularly difficult to acquire (Ciovarnache/Avram 2013; Ticio 2015). The issues learners face are assumed to derive from two main sources: the polyfunctionality of the marker *a* in Spanish and structural differences between the respective L1 and Spanish.

Focusing on the first source, the function of *a* as a differential marker of direct objects is not very transparent to learners. This is due to the fact that, in Spanish, *a* also functions as a marker of indirect objects (12a), serves as a locative preposition (12b), which relates to *a* being used to express the periphrastic future tense in a grammaticalized manner (12c).

(12) a. Juan le da un libro a Ana.
Juan her gives a book$_{\text{DO}}$ to Ana$_{\text{IO}}$
'Juan gives Ana a book.'
b. Juan y Ana van a un concierto.
Juan and Ana go to$_{\text{PREP.LOC}}$ a concert
'Juan and Ana are going to a concert.'
c. Juan y Ana van a ir a un concierto.
Juan and Ana go to$_{\text{PREP.FUT}}$ go to$_{\text{PREP.LOC}}$ a concert
'Juan and Ana are going to go to a concert.' (cf. Pomino 2012: 4)

Secondly, there is a well-known contrast between German, our L1 of interest, and other languages when it comes to distinguishing objects receiving accusative case. In German, case marking is associated with the declension class of the noun, with the specific morphological form being applied across the board for all objects that fulfil the respective syntactic role of the argument (e.g., *den Jungen* as the accusative object). By contrast, the *a*-marking of the direct object in Spanish depends on several syntactic, semantic, and pragmatic factors (cf. example 13).

(13)	a.	Sp.	El	hombre	ve	al	niño.	
			the	man	sees	DOM-the	boy$_{DO}$	
			'The man sees the boy.'					
	b.	Ger.	Der	Mann	sieht	den	Jungen.	
			the	man	sees	the	boy.ACC	
			'The man sees the boy.'					
	c.	Sp.	El	hombre	ve	el coche.		
			The	man	sees	the car$_{DO}$		
			'The man sees the car.'					
	d.	Ger.	Der	Mann	sieht	den	Wagen.	
			the	man	sees	the	car.ACC	
			'The man sees the car.'					(Pomino et al. 2018: 1)

Thus, Spanish DOM is likely to seem less comprehensive to L1 German speakers who are more familiar – be it explicitly or implicitly – with morphological case-marking than speakers whose L1 also features a DOM system.

Studies show that even highly proficient learners struggle to use DOM in a target-like manner. The results of Guijarro-Fuentes (2011) suggest that L2 learners (English as L1) acquire the distribution of DOM in a gradual way, starting from only one interpretable feature, animacy. In their analysis of three L2 corpora of Romanian and Spanish, Ticio/Avram (2015) observe that the acquisition process of DOM builds on the semantic features of animacy and specificity, which further aligns with Guijarro-Fuentes (2011) and Mayer/Sánchez (2021), who identify animacy as the most influential factor in the acquisition process of DOM for L2 learners. Only few studies have investigated how Spanish DOM is acquired by speakers with an L1-German background, with most of them focusing exclusively on heritage speakers (Pomino et al. 2018), leaving speakers who acquire Spanish as an L2 in a formal environment understudied. The majority of those studies built on experimental and elicited data, with the number of participants rarely above n = 15. To my knowledge, no corpus-based study has thus far addressed how L1-German speakers apply DOM in their L2 Spanish, much less through the analysis of corpus data with material from over 80 learners, which is the proposition of this study.

3. Interim summary and research questions

So far, Section 2 has reviewed the most prominent characteristics of Spanish DOM. All in all, the general underlying concept seems to be that the more individuated and prominent a direct object is, whether it be semantically or pragmatically, and the more it is affected by the predicate it complements, the likelier it is to be marked. Concerning challenges learners with an L1-German background may face in acquiring DOM, attention was drawn to the typological differences between Spanish and German, with the latter being characterized as lacking a DOM system. The research questions (RQ) of my study are the following:

RQ1 Do L1-German speakers apply DOM in their L2-Spanish differently compared to native speakers?

And, if the answer to this question is affirmative:

RQ2 In what ways do the L2 speakers differ from the L1-speakers in their use of DOM?

4. Data and methods

To answer the presented research questions, this section provides details on the data and the methods used for the analysis. Section 4.1 provides some information on the data, annotation, and speakers. Section 4.2 outlines the applied statistical methods.

4.1 Data and annotation

The chosen corpus for this study is CEDEL2 (*Corpus escrito del español como L2*, Lozano 2022), an open-access learner corpus of mostly written but also spoken vernacular Spanish. This corpus currently amounts to a total of 1,105,936 words from 4,399 different participants, which renders it one of the largest corpora of its kind (Lozano 2022). CEDEL2 was designed with a second-language acquisition agenda in mind and is primarily a corpus of L2 Spanish. However, additional data from speakers with Spanish and Latin American varieties as L1 have been collected for comparative purposes.

For this study, the written components of "Task 14: Chaplin" were used. This L2 Spanish subcorpus consists of about 16,100 words in the form of written texts from 82 L1 German speakers whose ages range from 19 to 71. The majority of L2 speakers are fairly proficient speakers of Spanish, with over 52% of participants scoring as advanced and only 6% scoring as beginners in the proficiency tests performed prior to testing. Due to this imbalance in proficiency, learners were treated as one group without further differentiation between the levels of proficiency. The collection of data was conducted at different universities or other educational institutions in Europe and achieved via the following task given to participants:

> Mira el siguiente vídeo de Charles Chaplin (4 min). Haz un resumen de la historia. Puedes ver el vídeo más de una vez.
> 'Watch the following Chaplin video clip (4 min) and retell the story. Summarize the story. You can watch the video clip more than once.'[2]

This comparatively narrow frame of a written task ensures a high degree of comparability between speakers' texts. Comparability is also maintained when it comes to the native reference corpus since only parts of the same task environment were considered for the analysis. Out of the 349 Task-14 texts, 20 texts were selected, amounting to approximately 4,500 words in the reference corpus. Due to this comparatively small amount of L1 texts, special attention was paid to the linguistic background of the native speakers. To avoid differences based on geographical variation and variation related to bilingualism, a range of different places of data collection were considered, including the south of Spain (Almería, Granada, Cádiz), central Spain (Madrid), northern Spain (Bilbao, Zaragoza), and the east coast (Barcelona, Valencia). Since several official languages are spoken in these areas and many speakers living there are bilingual, only participants whose L1 is Spanish and who speak Spanish at home were selected. The ages of these speakers extend from 21 to 56, with a majority of them ranging between 20 and 30, as is the case for the L2 subcorpus.

In an effort to minimize misclassifications, both the 20 L1 texts and the 82 L2 texts were manually checked and annotated for the respective variables.[3] First, negative hits (e.g., ditransitives, object pronouns) were separated from tokens eligible for DOM such as nouns, personal pronouns, and relative pronouns. The 219 remaining hits of the L1 corpus and the 868 positive hits of the L2 corpus were then annotated according to the variables listed in Table 2.

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2 The video (*Charlie Chaplin: The Kid*) can be found here: https://www.youtube.com/watch?v=eO1HvF2G2Sw.

3 CEDEL2 is POS-tagged with FreeLing (Padró/Stalinovski 2012). Relying on simple and advanced search was, however, not a feasible option for my study due to two main reasons: First, FreeLing lacks a means of differentiation between cases (except for pronouns) and can therefore not be used to extract direct objects from the data. Second, there is a high risk of misclassifications due to pseudo or novel words (e.g., *ve a una *mulier* 'he sees a woman'), misspellings (e.g., **encomtrar*), and grammatical mistakes (e.g., a learner using *me* 'me' instead of *mi* 'my') in the L2 corpus.

DOM	MARKED, UNMARKED
ANIMACY	ANI (animate), INANI (inanimate)
SPECIFICITY	SPEC (specific), UNSPEC (unspecific)
GIVENNESS	GI (given), NE (new)
CONTINUITY	CON (continuous), DIS (discontinuous)
TELICITY	TEL (telic), ATEL (atelic)
AFFECTEDNESS	HA (highly affected), A (affected), UA (unaffected)

Table 2. Overview of variables used in the annotation of the data and their respective levels.

While the annotation of ANIMACY is quite straight-forward, the classification of SPECIFICITY may require more elaboration. Out of the different types of determiners, numerals, definite determiners, and possessive determiners were categorized as SPEC. The use of indefinite determiners led to an annotation of UNSPEC, as did the absence of determiners or the use of an indefinite pronoun as the direct object. With regard to the information-structural variables, GIVENNESS was annotated by means of two levels: NE ("new") and GI ("given") instead of topicality. Annotating written texts which are expressed in the 3rd person perspective according to discourse-dependent categories is not only difficult, but the many different definitions surrounding the term *topicality* also bare a high risk of intransparency. These issues motivated the decision to focus on a specific characteristic of topicality, GIVENNESS, as a more transparent variable.

Another information-structural variable which partially overlaps with the distinction between presupposed and new referents is (object) CONTINUITY, with the two levels CON ("continuous") and DIS ("discontinuous"). The transitivity-related variables were annotated with consideration of the whole clause or sentence, including potential adverbial adjuncts. For TELICITY, two values TEL ("telic") and ATEL ("atelic") were used since there is no clear definition as to when a verb is neither atelic, nor telic. As for AFFECTEDNESS, three levels were annotated to establish a range of degrees of affectedness in the verb-direct object combinations found in the data, following Tsunoda's (1985) and Hopper/Thompson's (1980) classification groups: highly affected (HA), affected (A), and unaffected (UA). The final variable to be annotated was DOM, relating to the presence or absence of the marker *a* with the respective direct objects. For a better insight into the data of both subcorpora, Tables 3 and 4 display the absolute and relative frequencies of the direct objects considered in the analysis according to the above-mentioned variables.

L1 Spanish						
variable	**level**	**frq #**	**frq %**	**level**	**frq #**	**frq %**
DOM	marked	101	0.46	unmarked	118	0.54
ANIMACY	animate	113	0.52	inanimate	106	0.48
SPECIFICITY	specific	133	0.61	unspecific	86	0.39
GIVENNESS	given	94	0.43	new	125	0.57
CONTINUITY	continuous	17	0.08	discontinuous	202	0.92
TELICITY	telic	129	0.59	atelic	90	0.41
AFFECTEDNESS	affected, highly affected	191	0.87	unaffected	28	0.13
TOTAL		219				1.0

Table 3. Overview of token numbers according to variables in the L1 Spanish corpus.

L2 Spanish						
variable	**level**	**frq #**	**frq %**	**level**	**frq #**	**frq %**
DOM	marked	217	0.25	unmarked	651	0.75
ANIMACY	animate	605	0.70	inanimate	263	0.30
SPECIFICITY	specific	550	0.63	unspecific	318	0.37
GIVENNESS	given	463	0.53	new	405	0.47
CONTINUITY	continuous	201	0.23	discontinuous	667	0.76
TELICITY	telic	574	0.66	atelic	294	0.34
AFFECTEDNESS	affected, highly affected	714	0.82	unaffected	154	0.18
TOTAL		868				1.0

Table 4. Overview of token numbers according to variables in the L2 Spanish corpus.

4.2 Statistical methods

Second language research increasingly uses empirical approaches to address psycholinguistic questions in areas such as processing, sentence complexity, and cross-linguistic transfer (Deshors/Gries 2023: 164). The most widely used statistical tool in this respect has probably been that of regression modeling (Gries 2020: 618). Despite of the unquestioned usefulness of logistic regression models, other methods may be more helpful and effective in specific cases where the composition of the data creates difficulties in the application of parametric regression modeling. A first assessment of the data of this study within a regression analysis returned high values of multicollinearity (variance inflation factor (vif) > 7000 for some of the predictors). With regard to some of the predictors, some confusion matrices contained empty cells, pointing to the data being quite unbalanced (Gries 2020: 618), which is – as Tagliamonte/Baayen (2012: 172) point out – typical of natural linguistic data. However, given that these conditions are highly challenging for regression modeling, potentially leading to an unreliable assessment of the strength of a given predictor, the use of a different statistical approach seemed to be advisable. The use of random forests has become a particularly popular alternative in recent years (Gries 2020: 618). Random forests, developed by Breimann (2001), are ensembles of decision trees, whereby randomly sampled data is used to arrive at an aggregated estimate of a particular outcome's probability through random permutation tests (Breimann 2001: 113; Strobl et al. 2009: 325).

It is a partitioning approach that consists of hundreds of different trees in which the data is repeatedly categorized into binary splits (Deshors/Gries 2016: 202). The associations of a particular predictor are hereby tested by working through the data and by a trial-and-error procedure in the form of singular voting (Tagliamonte/Baayen 2012: 159). To evaluate how useful a predictor is, a permutation-variable importance measure is used. This measure is acquired through permuting the values of the predictor and breaking its association with the response variable (Tagliamonte/Baayen 2012: 160). If the model's performance decreases considerably, then the predictor must be relevant (Breimann 2001: 115). The variable importance is generally described as the difference in prediction accuracy before and after permuting the predictor (Tagliamonte/Baayen 2012: 160; Strobl et al. 2007: 29). Based on recent findings by Debeer/Strobl (2020: 318) on the accuracy of different kinds of variable importance measures, in this study, Conditional Permutation Importance (CPI) will be preferred over the more commonly used Gini-Index.[4]

Due to their non-parametric nature, random forests are assumed to be much less affected by data sparsity, collinearity, and overfitting (Breimann 2001: 113; Gries 2022: 618). To prevent overfitting, each tree in a random forest is grown for a subset of the data generated by randomly sampling without replacement from observations and predictors. Consequently, for each tree a training set is paired with a test set using the remaining data (out-of-bag observations, OOB) (Tagliamonte/Baayen 2012: 159, 161). Random forests may therefore be able to provide models that are more suitable for generalization than regressions, where the

[4] The Gini Index is an entropic variable importance measure used to quantify the impurity of each node that is implemented in the *randomForest* package. This measure has, however, been found to be biased, as pointed out by Strobl et al. (2009). Instead, they recommend using the Conditional Permutation Importance measure (CPI), which randomly permutes the values of a predictor, thus breaking its original association with the response variable. Therefore, the CPI was the measure of choice for this study and implemented using the package created by Debeer et al. (2021).

accuracy of the model commonly increases with every parameter added to the model. Random forests are also more flexible with regard to highly unbalanced designs and may therefore yield superior models (Tagliamonte/Baayen 2012: 135).

Despite the many advantages of random forests, some shortcomings exist, which mainly relate to the difficult visualization and interpretation of the end result (Deshors/Gries 2016: 203). This is due to the absence of an obvious way to compute *p*-values for predictors since random forests consist of hundreds of different trees. As a means of visualizing the random forest and simultaneously using a classification method that includes *p*-values, the interaction of predictors is often shown with the use of a conditional inference tree. This practice is often criticized as "misrepresent[ing] the results of the random forest" due to a high degree of variability (Gries 2020: 637). As a better solution, a representative tree (*reprtree( )* function; Banerje et al. 2012; Dasgupta 2014) should be used, whereby the most representative tree of the random forest is identified and selected.

In sum, while individual decision trees provide results that seem relatively intuitively interpretable and visualize how the predictors operate in tandem, their isolated use is not recommended. To obtain a more reliable idea of predictor importance, only representative trees of the random forest should be presented in complementation to the evaluation of variable importance. With these limitations in mind, random forests offer a useful alternative to traditional regression modeling.[5]

For this study, a fairly new approach will be employed, referred to as MuPDARF[6] developed by Deshors/Gries (2016), in which two models are created using random forests. The point of this approach compared to a traditional regression-only kind of analysis is to explore deviances between reference and target speakers. A regression analysis might return significant differences between the levels of a predictor even if they do not actually lead to different categorical decisions (Gries 2022: 260), MuPDARF however focuses exactly on those cases.

5 In the present analysis, random forests were generated using the *randomForest( )* function in R's *randomForest* package (Hothorn et al. 2006; Strobl et al. 2007). The variance importance measure that was applied is CPI, which is included in the *permimp* package (Debeer/Strobl 2020: 318). The tree plots were created using the *reprtree* package.

6 MuPDARF is a revised version of the original MuPDAR approach (Multifactorial Prediction and Deviation Analysis using Regressions) developed by Deshors/Gries (2014).

MuPDARF consists of the following four steps: First, one fits a first random forest model (R1) on the reference speakers' (L1-Spanish speakers) choice of using DOM (*ntree* = 2000, number of trees used in aggregation). This model contains DOM as the response variable, and ANIMACY, SPECIFICITY, GIVENNESS, CONTINUITY, TELICITY, and AFFECTEDNESS as predictors (Gries/Wulff 2021: 282). Secondly, the goodness of fit of the model is checked. If the model is sufficiently accurate, it is then applied to the target speaker group (L2-Spanish speakers) in a third step to impute for every target speaker choice what a reference speaker would have done in the same situation (Gries/Deshors 2020: 85). The result of this step is the addition of a new factor with the values nativelike/non-nativelike or MATCH/NO MATCH. As a last step, one then explores the degree to which the target speaker made canonical choices by virtue of a second random forest model which involves only the L2-speaker data. In this model, the newly added factor then provides the response variable REFERENCE-LIKENESS and contains DOM as a predictor in addition to the predictors of the first model (Gries 2022: 257).

5. Results

As explained in Section 4, the statistical analysis of this study consists of the so-called MuPDARF approach, which involves a series of steps. Therefore, in this section, the results will be reported in a way that reflects the sequence of these steps.

5. 1 Fitting of the first random forest on native speaker data

The first random forest (RF1) was created using DOM as the response variable (reference level "unmarked") and ANIMACY, SPECIFICITY, GIVENNESS, CONTINUITY, and AFFECTEDNESS as predictors (*ntree* = 2000, *mtry* = 3). Since this first random forest was a reference model, it was fit on the native speakers' data by separating the 868 L2 tokens from the 1087 tokens of the whole data frame. The confusion matrix shown in Table 5 indicates that the model correctly predicted all observations of unmarked tokens, and falsely predicted 12 unmarked tokens as marked.

	unmarked	marked
predicted as unmarked	106	12
predicted as marked	0	101

Table 5. Confusion matrix of the predictions of the first random forest (RF1) for DOM with the model's predictions in the rows and the actual outcome in the columns.

Thus, the first analysis returned a classification accuracy of 94.5% and an OOB-error estimate of 5.48%. This can be interpreted as a good fit considering the assumed threshold value of 80% for most goodness-of-fit estimates. Since the model was good enough, it could be subjected to further examination such as determining the importance of each predictor with regard to the prediction of DOM. Figure 1 presents the variable importance in the form of the CPI for all of the included predictors. The grey vertical line highlights the variable importance of the inconsequential predictors, which is equal to 0. Returned values below zero and values varying randomly around zero are interpreted as showing no significant influence (Tagliamonte/Baayen 2012: 23).

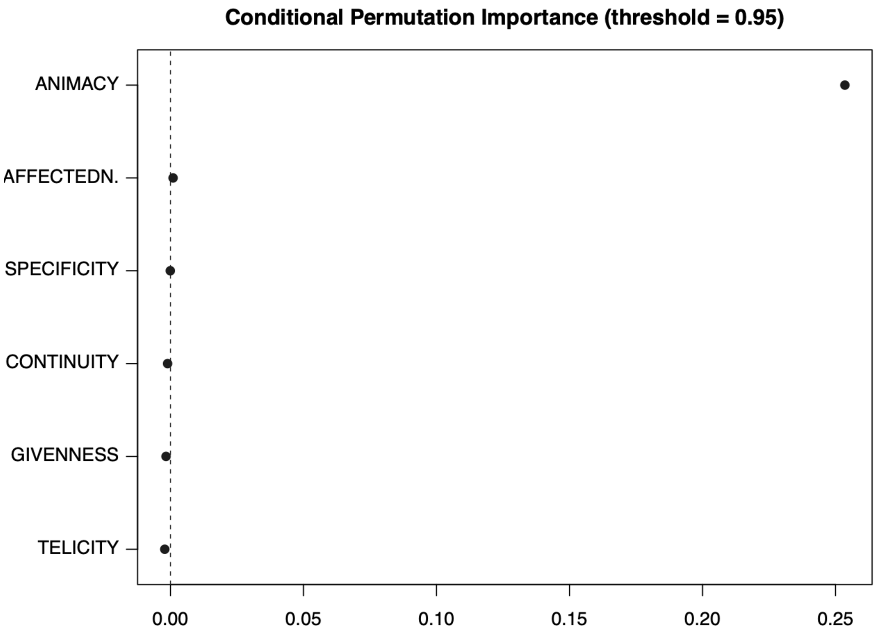

Figure 1. Variable importance measure of RF1 using CPI.

In this case, ANIMACY, SPECIFICITY, and AFFECTEDNESS showed values greater than zero, while GIVENNESS, CONTINUITY and TELICITY received values below zero. Comparing the predictors which appear as influential, ANIMACY is by far the most important predictor, which is in line with the majority of assessments of the influence of animacy on DOM. With a certain distance in the x-value zero, AFFECTEDNESS and SPECIFICITY can be considered as marginally influential. None of the other predictors contribute statistically significant effects, as indicated by the vertical grey line. Note that, since all associations with other predictors were broken, this plot shows all predictors as considered in their own right. Interactions between predictors are therefore included in the measure of variable importance and some of the other predictors may have contributed to the importance of ANIMACY. Since potential interactions cannot be seen in a variable importance plot, let us look at a representative tree of this random forest as a means of visualizing how the predictors interact in tandem. The selected tree for the RF1 model can be seen in Figure 2.

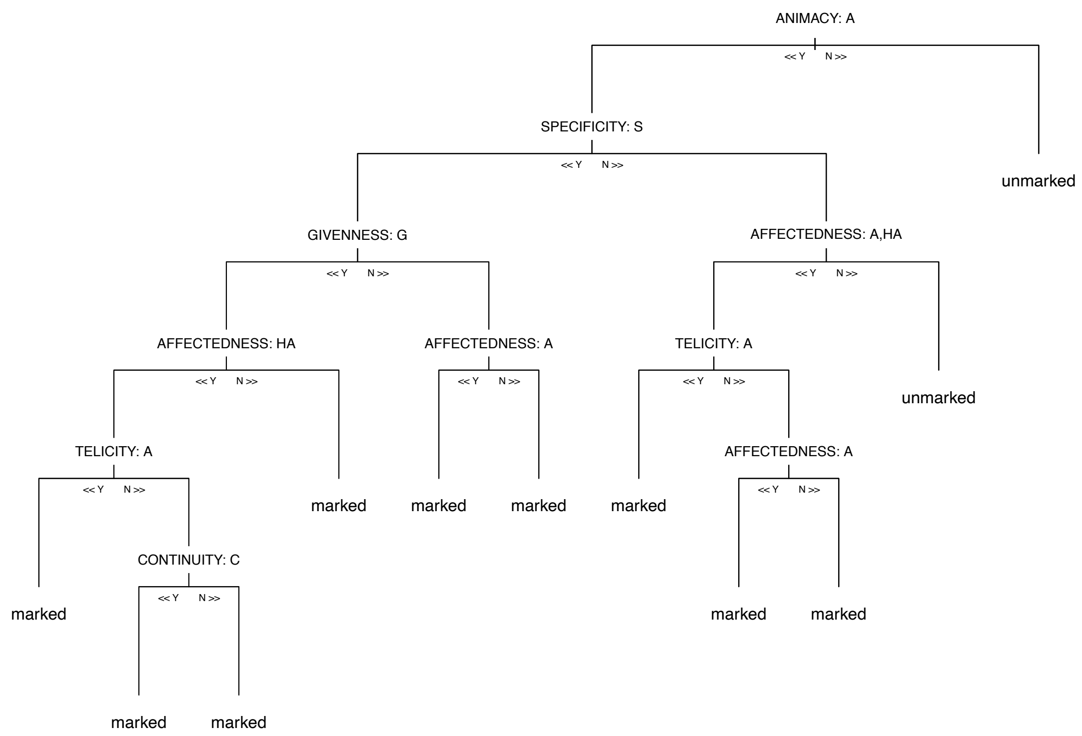

Figure 2. Representative tree of the random forest RF1.

The classification tree resulting from the native speaker data can be interpreted as a sequence of conditions and their consequences. The splits are based on a specific factor level, which consists in the respective reference levels of each predictor. These levels are indicated by abbreviations in Figure 2, "N" meaning 'no' and "Y" meaning 'yes'. Starting from the top, if the direct object is not animate, there is only one possible outcome, which is being left unmarked (14a). If the direct object is animate, additional other predictors play a role in the outcome of *a*/∅-alternation. Continuing with the splits on the left, if the animate direct object is specific, all of the subsequent splits share the same outcome of the direct object being *a*-marked. This includes specific direct objects that had already been introduced at an earlier point in time (14b) and new direct objects (14c). On the other hand, if the direct object is unspecific, AFFECTEDNESS determines whether the direct object is marked or not, since out of these unspecific tokens, only those which are affected or highly affected are marked (14d), whereas unaffected ones are left unmarked (14e). The latter is associated with the characteristics of DOM with verbs such as *tener*.

(14) a. y descubre una nota bajo la ropa del bebé (L1, T6 ,17)
and finds a note$_{DO}$ under the clothes of-the baby
'and he finds a note under the clothes of the baby'

b. al reconocer a Chaplin (L1, T15, 61)
to recognize DOM Chaplin
'when he recognizes Chaplin'

c. En una esquina, encuentra a un bebé abandonado (L1, T06, 05)
at a corner, finds DOM a baby$_{DO}$ left
'at a corner, he finds a baby that was left behind'

d. y consigue engañar a un señor (L1, T16, 23)
and manages fool DOM a man$_{DO}$
'and he manages to fool a man'

e. y que ya tiene otro bebé (L1, T07, 07)
and who already has other baby$_{DO}$
'and who already has another baby'

Returning to the left side of the second split above, at the node of the predictor GIVENNESS, we can see that several splits occur in the case of the animate DO being given. Note, however, that all of the possible outcomes of these splits are the same, namely that the DO shows a tendency of being *a*-marked. Having accounted for the first model's goodness-of-fit and checked a representative tree of the random forest, it could be used for the next step.

5.2 Applying the first model to the learner data

In this step, the RF1 model was used to compute a random forests-based prediction for every case in the L2-Spanish data, using the generic *predict( )* function of R. Note here that the PROFICIENCY of learners will not appear as a predictor in the following steps after having originally been included as a factor in the first round of analyses and shown no influence on DOM. This is most likely due to the data being skewed in terms of proficiency. Given the high risk of inconclusive results, proficiency was eventually excluded as a predictor from the present analysis.

As can be inferred from Table 6, the RF1 model falsely predicted 380 tokens as unmarked when the actual observations were in fact marked, and falsely predicted 3 tokens as marked which were unmarked in the actual data. Due to this increase in misclassifications, the overall classification accuracy decreased to 55.9%. However, this is not surprising since the L2 speakers are expected to deviate from the native speaker data.

	unmarked	marked
predicted as unmarked	271	3
predicted as marked	380	214

Table 6. Confusion matrix of the predictions of the RF1 model for DOM in the L2-data with the model's predictions in the rows and the actual outcome in the columns.

This step therefore confirmed the alternative hypothesis behind the research question (A), which assumed a difference between the use of DOM between native speakers and L1-German learners of Spanish. The following steps were thus used to identify the ways in which the learners' use of DOM differed from that of the native speakers.

5.3 Second random forest on differences between native speakers and learners

To identify the differences between the native speakers' and the learners' choices, an additional factor, REFERENCE-LIKENESS, with the levels MATCH and NO MATCH was computed to encode whether the learners' choices differed from the native speaker predictions. REFERENCE-LIKENESS was therefore modeled as a function of the same predictors as used before. The former response variable DOM became one of the predictors to determine to which degree DOM would be able to account for the deviances between natives and learners. The overall results were fairly good in the sense that the statistical model was able to predict the REFERENCE-LIKENESS very well.

	MATCH	NO MATCH
predicted as MATCH	485	0
predicted as NO MATCH	3	380

Table 7. Confusion matrix of the predictions of the RF2 model for REFERENCE-LIKENESS with the model's predictions in the rows and the actual outcome in the columns.

The confusion matrix in Table 7 lists the predictions of the RF2 model, out of which only three tokens were misclassified, resulting in an excellent classification accuracy of 99.4% and an OOB error estimate of 0.35%. Considering the model's goodness-of-fit, the results could be explored further by assessing the importance of individual variables.

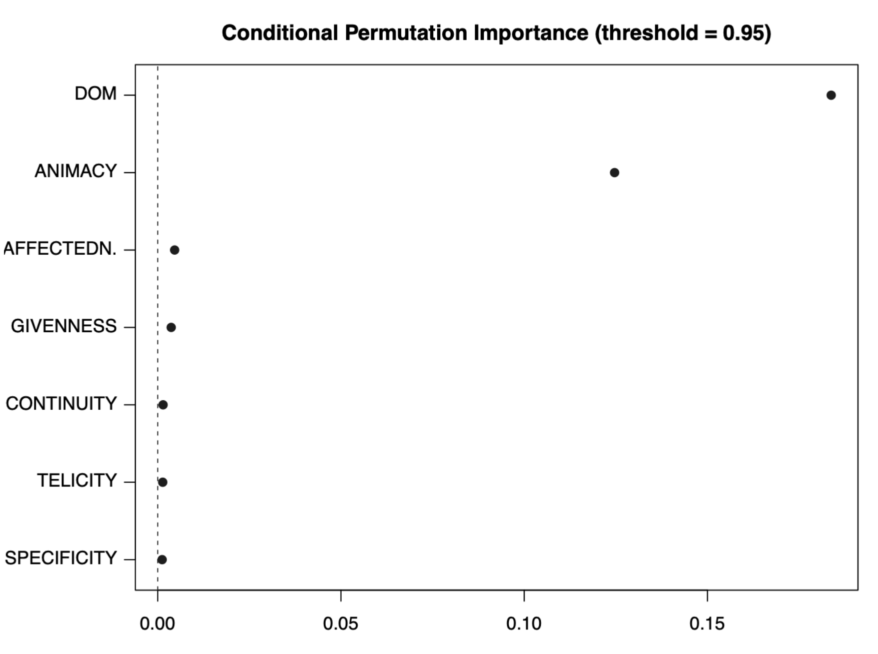

Figure 3. Variable importance measure of RF2 using CPI.

The variable importance plot for the RF2 model in Figure 3 shows that DOM, listed as a predictor now, was the most influential factor in accounting for the deviances between native speakers and learners, followed by ANIMACY. In this analysis, and contrary to the previous random forest, all predictors showed some degree of influence. Here, AFFECTEDNESS and GIVENNESS are depicted as more important than TELICITY, SPECIFICITY and CONTINUITY, which can be interpreted as marginally influential judging from their close position to the grey line. This analysis can be interpreted as a second confirmation of the alternative hypothesis of RQ1, which states that there is a difference between native and non-native use of DOM. As a means of getting a better idea of the ways in which the predictors lead to different outcomes in *a*-marking in the two groups, let us look at the respective representative tree in Figure 4, below. The overall structure of the representative tree implies that learners generally marked less direct objects than natives since the majority of the NO MATCH outcomes are created from the node where the predictor DOM is represented with the reference level UNMARKED (indicated as "u" at the top node). Starting again from the top, it appears as though the DOs that were marked by learners would have also been marked by natives as can be seen on the right side of the first split which indicates a match (15a). This means that learners perform completely target-like with regard to their a-marked direct objects.

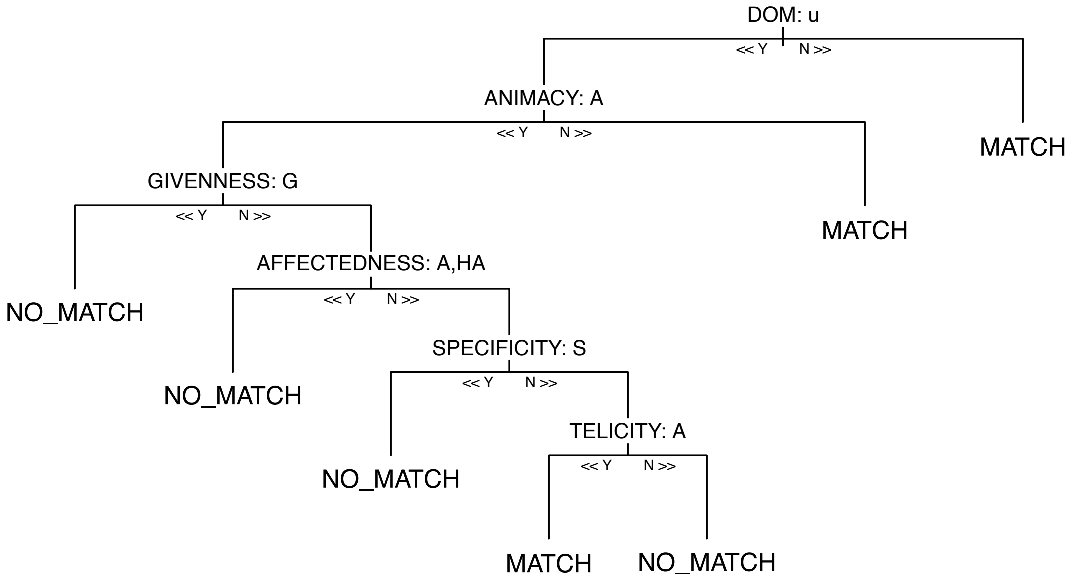

Figure 4. Representative tree of the random forest RF2.

In the case of unmarked DOs, however, learners performed somewhat differently to natives. This difference manifests itself through various splits involving most of the predictors from above and reflecting the importance of the predictors in the hierarchy of the tree's structure. As long as the DOs in question were inanimate, learners showed the same behavior as the reference speakers by overall not marking inanimate tokens (15b). Regarding animate direct objects, the only example of target-like performance of DOM was if the animate object was not given in the context, unaffected, unspecific, and atelic. For these instances, both natives and learners left the DO unmarked (15c).

For the remaining cases of animate direct objects, the analysis returned a lack of reference-likeness in the case of animate objects given in the context (15d), for objects classified as all of the just mentioned in addition to being affected or highly affected (15e), for objects classified as all of the just mentioned in addition to being specific (15f), and for objects with all of the just mentioned classifications in addition to being telic (15g). For these cases, learners chose to leave the direct object unmarked where native speakers would in turn have made the choice to apply DOM.

(15)[7] a.	que	(…)	quiere	poner	al	bebé	en	el	suelo	(L2, T20, 18)
	who		wants	put	DOM-the	baby$_{DO}$	on	the	ground	

‘who, (…), wants to lay the baby on the ground.’

[7] As this study is not concerned with the acquisition of orthography, any spelling mistakes in the learners' written production data were corrected in the examples. Lexical errors such as

b. un hombre tira basura de una ventana (L2, T10, 2)
a man throws trash$_{DO}$ off a window
'a man throws trash out of a window'

c. que necesita alguien (L2, 56, 37)
who needs someone$_{DO}$
'who needs someone'

d. Pero esta persona ve el paseante de la mujer también. (L2, T20, 27)
but this person sees the pram$_{DO}$ of the woman also
'But this person sees the pram of the woman as well.'

e. cuando encuentra otro hombre (T69, 19)
when finds other man$_{DO}$
'when he finds another man'

f. Luego Charlie toma el bebé de nuevo. (L2, T62, 27)
then Charlie takes the baby$_{DO}$ again
'Then, Charlie picks up the baby again.'

g. y acuesta el bebé en el vagón (L2, T80, 19)
and lays the baby$_{DO}$ in the pram
'and lays the baby down in the pram'

The second random forest therefore also provides an answer to the second research question, which was directed at the ways in which the L1-German speakers differed from the native speaker group in their use of Spanish DOM.

6. Discussion

In the following, a short summary of the key findings will be provided, followed by the discussion of the results in light of earlier research on second-language acquisition of DOM and potential implications for the L2 instruction of DOM.

6.1 Main findings

Starting with the key findings, let us recall the two main research questions presented in Section 3:

RQ1 Do L1-German speakers apply DOM in their L2-Spanish differently compared to native speakers?

And, if the answer to this question is affirmative:

RQ2 In what ways do the L2 speakers differ from the L1 speakers in their use of DOM?

In response to RQ1, the essential step was to model a random forest classifier (RF1) on the native speaker data. The decrease in accuracy from 99.4% to 55.9%

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paseante 'passerby' (15d) instead of *carrito (de niño)* 'pram' are rendered in glossing and translation with the respective intended meanings.

when used on the L2-data was a clear indicator of major deviances between natives and learners in their use of DOM. This observation aligns with the majority of studies on L2 usage of DOM, including Guijarro-Fuentes (2011), Mayer/Sánchez (2021), and Simon (2022) conducted on L2 data with L1 backgrounds other than German.

The second research question (RQ2), which was directed at the ways in which L2 speakers differed from L1 speakers in their use of DOM, was answered with the creation of a second random forest model (RF2) with REFERENCE-LIKENESS as the new response variable and DOM as one of the predictors. This model's accuracy was excellent and returned some valuable insights. The variable importance assessment showed that DOM was in fact responsible for the majority of differences between the L1 and the L2 data, which can be interpreted as a second confirmation of the first alternative hypothesis behind RQ2. Another highly influential predictor was ANIMACY, followed by the marginally influential predictors AFFECTEDNESS and GIVENNESS. With regard to the specific differences, the first observation was that the overall structure of the representative tree implied that learners generally marked less direct objects than natives, since the majority of the NO MATCH outcomes are created from the node where the predictor DOM is represented with the reference level UNMARKED.

Focusing first on the cases in which learners performed in a target-like manner, all of these cases were instances where natives would have chosen to *a*-mark the direct object. This implies that learners performed completely reference-like with regard to their *a*-marked direct objects. As for the deviating cases, learners chose to leave a considerable number of animate direct objects unmarked that would have been marked by natives. This can be inferred from the fact that all of the NO MATCH outcomes spawn from the path UNMARKED x ANIMATE. Here is where ANIMACY comes into play as the key predictor of reference-likeness since it is able to account for the target-like and non-target-like behavior of the learners. In sum, these observations provided the answer to RQ2. The decisive property of ANIMACY as a predictor of DOM was also found in the above-mentioned empirical studies by Ticio/Avram (2015) and Parodi/Avram (2018). Given that these studies featured Romanian, English, and German as L1 of the learners, it appears as though ANIMACY maintains its influence on DOM regardless of the learners' L1. In sum, to increase reference-likeness, learners would have to mark more of their animate direct objects. The cases in which they choose to apply DOM, however, tend to be target-like instances of DOM, where natives would have behaved in a similar way.

This implies that learners may have a general understanding of DOM but likely have not acquired it to the point of automation. The delay in the acquisition of DOM compared to other properties of Spanish may be explained by potential negative L1 transfer or the lack of positive transfer from German, which lacks a DOM system (Schmitz 2006: 245). Moreover, considering that the omission of the marker only leads to misunderstandings in very few cases, speakers may pay less attention to DOM compared to other properties which they might regard as communicatively more important.

On another note, this study also provides some insight into the ways DOM is applied by native speakers of Spanish by examining the RF1 model and its values for variable importance. This model clearly shows that the majority of the outcomes of the response variable DOM could be accounted for by the predictor ANIMACY, which aligns with the consensus of researchers who have previously investigated the L1-usage of DOM. While qualitative analyses of DOM have brought forward a plethora of potential factors which are assumed to influence *a*/∅-alternation, there are no studies, to my knowledge, which empirically test all of these factors on contemporary Spanish as a way of determining their proportional influence on DOM. Since the analysis of this study included the most discussed semantic and information-structural factors as well as the most prevalent verb-related factors, the variable importance assessment of the RF1 model sheds new light on the proportional influence and interactions of these factors. The results indicated that animacy is by far the most influential predictor, with the others only displaying marginal to no significant importance. It should be noted, however, that this could be due to the small L1 dataset. The slight influence of other predictors such as givenness could be deducted from the path of the first representative tree, which indicated cases where animate objects were not marked if they were not given (i.e., NEW) in the context as well as unaffected, unspecific, and atelic. These cases include constructions with the verb *tener* 'have', which is generally considered to block DOM (cf. 16).

(16) Dice que el bebé no tiene padres. (L2, T22, 21)
says that the baby not has $\text{parents}_{\text{DO}}$
'It says that the baby doesn't have parents.'

It should be noted, however, that in order to confirm these proportions, more data has to be tested, specifically data involving other registers, both in spoken and written form, since the data of this study was quite particular in being written in a 3rd person narrative/descriptive style.

6.2 Possible implications for L2 instruction of Spanish

Even though DOM is considered as a particularly vulnerable domain in the second language acquisition of Spanish, its presence in curricula used for the instruction of Spanish as a foreign language in Germany is sparse. Checking the ten most commonly used textbooks and companions, only two, i.e., *¡Apúntate!* (Balser 2009) and *Línea verde* (Bade et al. 2006) pay explicit attention to DOM by incorporating explanations, exercises, and tips in the vocabulary lists. In contrast, DOM is neglected in the remaining textbooks, such as *Caminos nuevo* (Görissen et al. 2010), *Contigo* (Duncker/Hammer 2010), and *Encuentros* (Marín Barrera et al. 2010). In the textbooks where descriptions are included, they are highly reduced. In this sense, the partial acquisition of DOM in learners is a direct reflection of its representation in materials and curricula.

It was emphasized that the deficiencies are not so much in the fundamental understanding of DOM but primarily in the lack of awareness in both spoken and written language use. Many learners appear to not have achieved automation yet, which results in frequent omissions of DOM. To increase the necessary automation regarding DOM in the Spanish of learners, existing implicit techniques may be expanded, and new techniques may be developed. The highlighting in vocabulary lists can be maintained and enhanced by, for instance, increased representation in word chunks (i.e., listing intransitive and transitive structures as collocations in the vocabulary list). This could lead to stronger contextualization and cognitive processing in a pragma-linguistic sense. Furthermore, it may be effective to emphasize DOM iteratively throughout the textbook instead of a one-time mention of this grammatical phenomenon during five consecutive years of a curriculum. This could be achieved through visual highlighting in the form of underlining, marking, and commenting on DOM structures in lesson texts and assignments. A recurrent exposure to DOM and familiar verb patterns would reactivate stored information about transitive sentences with animate objects, which could also lead to improved memorization and easier retrieval (Fäcke 2011: 158–165). The effectiveness of changes in methodology such as the suggestions made above still has to be investigated.

7. Conclusion

The present quantitative study on the use of DOM in the L2 Spanish of L1-German speakers offers new empirical evidence for the discussion on Spanish DOM and its acquisition from an L1-German perspective. The main points addressed in this paper concern the difference between L1 and L2 usage of Spanish DOM, testing whether there are differences between natives and learners and, if the answer to this question was affirmative, in what ways L2 speakers deviated from native speakers.

For the analysis, an approach called MuPDARF (Deshors/Gries 2016) was applied, which consists of a multi-step analysis featuring a random forest modeled on native speaker data and a second random forest modeled on the criterion of reference-likeness to directly compare natives' and learners' choices in comparable contexts. Overall, the results indicated that learners are capable of using DOM in a target-like manner but omit a considerable number of markers in instances where natives would have *a*-marked the direct objects. To explain these findings, various suggestions were offered such as the general lack of awareness of DOM, the relative subtleness of the communicative difference the omission of the marker implies, and the highly reduced instruction of DOM in teaching Spanish as a foreign language.

To increase the necessary automation regarding DOM in the Spanish of learners, classroom and textbook-guided instruction of DOM may be innovated. While some suggestions for improvement were offered in this article, an extensive revaluation of the current subordinate role of DOM in school curricula and the effectiveness of changes in methodology may be necessary. As a way of informing potential innovations in the L2 instruction of DOM, further corpus-based studies on the L2 acquisition of DOM should investigate the use of DOM by L1-German speakers in additional genres and registers.

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